

The Utility Problem

Planned outages have been an integral part of the electricity distribution business since the beginning of the electricity distribution industry. An outage is a state in which no power is delivered to the end consumer of the electricity, which is also called a no voltage state. Compared to unplanned outages, which happen suddenly due to a malfunction of equipment or another type of failure, planned outages are outages which are scheduled and properly announced to affected electricity consumers. These planned outages occur mainly due to grid activities which are related to necessary work on electrical distribution equipment needing upgrades or exchanges (it is important to mention that another type of planned outage recently emerged in certain parts of the world where the planned outage aims to de-energize the distribution equipment in order to prevent forest wildfires).

When certain work needs to be performed on the grid, it often requires de-energizing the equipment. If the equipment's topology lies on the path of power delivery and there is no other path to deliver the power, the activity will require the no voltage state (an outage) to a certain part of the grid. The outage will remain in place until the work is done and the equipment is back in operation. With hundreds of work orders which are issued in utilities every day, many of the work orders lead to planned outages. Planned outages are classified by utilities as an unavoidable consequence of grid upgrades.

With the emergence of modern analytic approaches, the view on planned outages is changing. These analytic approaches provide new possibilities, making electric distribution operations more efficient. An example is a detailed analysis of the planned outages by the Asset Management (AM) department. AM analysts can analyze the grid assets which will be under planned outage and recommend the work or replacement of the equipment during the same planned outage time. An example case might be an upgrade of the feeder's lateral where the conductors should be replaced. This work will introduce the outage into the entire lateral. AM analysts can analyze the assets under planned outages and prioritize other multiple assets for replacement during the same outage, whereas work on these would mean another planned outage. With this approach, the amount of downtime (outages) decreases, which leads to improved outage indicators, financial savings and better satisfaction for electricity consumers.

The Data Solutions and Methods

For the data analyst to conduct this type of analysis, it is important to mention that the right datasets and right data access methods are necessary. For the datasets, it is necessary to have a synchronized, centralized and cleansed data layer where all required data are digestible modelled. The data layer should recognize the following data source types:

- Outage Management System (OMS) data or work order system data for information about upcoming planned outages;
- GIS data with connectivity information for evaluation of the outage-affected area;

- Asset information and asset maintenance data for evaluation of the asset upgrade candidates.

It is not just the datasets which are necessary. It is also important to have ways and methods to work with this data effectively.

Awesense Platform and its Energy Data Model (EDM) interfaces are the right tools which any analyst can use to conduct an Analysis of the Planned Outage for Asset Upgrades. The Awesense platform's ontological data model recognizes all required data sets which are used in this type of analysis. In addition, the EDM's APIs and functions offer simplified retrieval of the data such that the building of such analysis is maximally efficient.

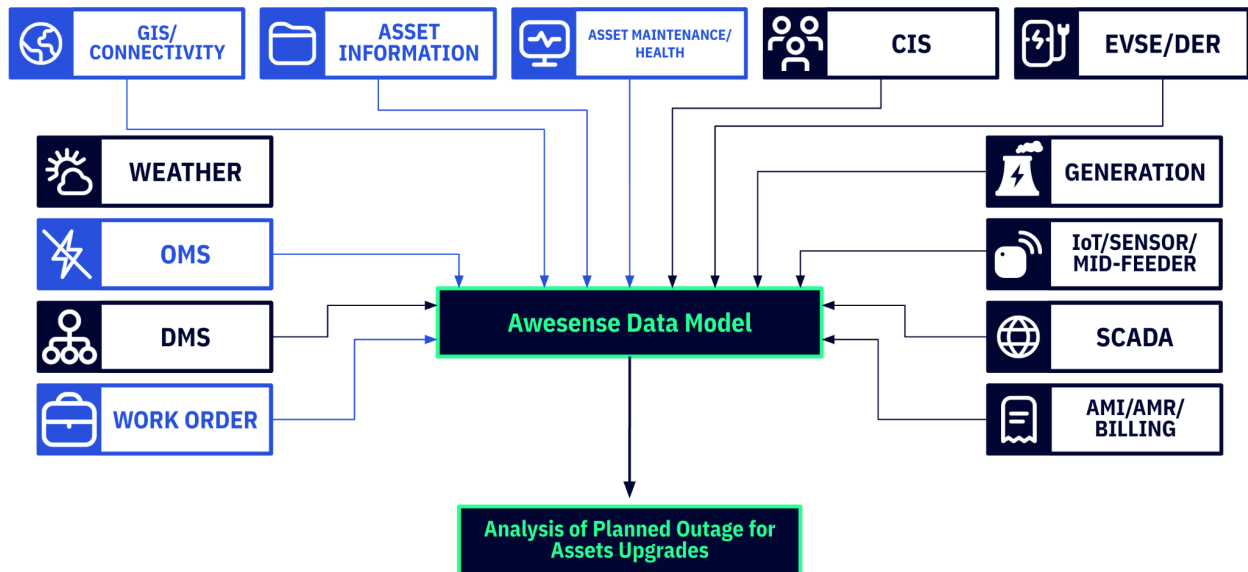
For the building of Analysis of Planned Outages for Asset Management on top of the Awesense platform, the data analyst should follow these steps:

- Obtain the information about upcoming planned outages along with the ID of the GIS element, which is the subject of work or upgrade;
- Use the GIS element ID and work with the GIS data to specify the outage-affected area, which will be under no voltage state. An analyst can use *grid_get_downstream()* or *grid_get_connected()* function for these purposes;
- After the specification of the outage-affected area, the analyst can target any asset group such as transformers, line segments, switchable elements, etc., for a detailed look;
- The targeted asset group has all asset management-related attributes like type, make, model, age, maintenance records, etc., available for better decision making;
- Choose one or multiple assets which can be worked on during the same planned outage time.

An above-mentioned process is just a high-level approach. However, the Awesense platform offers possibilities for a much more detailed approach.

Required Data Sets

The required data for this use case are visualized in the following diagram. Data sources highlighted in [royal blue](#) are required for this use case.



Available Use Case Implementations:

Users of the Awesense Platform can access this use case by the following means:

1. GitHub [link](#) to Jupyter notebook.
2. MSFT PowerBI file can be provided upon request.